

Notes: Barber and Odean (JF, 2000)

Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Household account data
- 2.2 Sample restrictions and household coverage
- 2.3 Trade-size and transaction-cost measures
- 2.4 Turnover
- 2.5 Gross household returns
- 2.6 Net household returns
- 2.7 Aggregate versus average household returns

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Own-benchmark abnormal return
- 3.2 Market-adjusted abnormal return
- 3.3 CAPM alpha
- 3.4 Fama–French three-factor alpha
- 3.5 Turnover-quintile test of overconfidence
- 3.6 Momentum-augmented specification
- 3.7 Taxable versus tax-deferred accounts

4 Identification and interpretation (what makes the estimates credible)

- 4.1 No instrument, but unusually direct measurement
- 4.2 Why the net-return result is hard to dismiss
- 4.3 Why own-benchmark returns are important
- 4.4 Why factor adjustments matter

4.5	Overconfidence versus rational information trading	
4.6	Alternative explanations	
4.7	Main limitation	
5	Tables and figures	
5.1	Figure 1: Monthly turnover and annual performance	
5.2	Table I: Trade size, transaction costs, and turnover	
5.3	Table II: Full-sample abnormal performance	
5.4	Table V: Turnover quintiles	
5.5	Table VI: Taxable and tax-deferred accounts	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** How well do individual investors perform when they hold and trade common stocks directly, and does active trading improve or damage their net returns?
- **Setting.** The paper studies 66,465 households with common-stock positions at a large U.S. discount brokerage firm from 1991 to 1996. The data contain monthly positions, trades, commissions, and transaction prices.
- **Core challenge.** To evaluate individual investors, the authors must separate stock-selection ability from transaction costs, risk exposure, portfolio style, and simple market movements.
- **Main conceptual contribution.** The paper turns administrative brokerage records into a household-level test of active trading. It asks whether the investors who trade most earn the higher gross returns required by rational information-based trading, or whether they simply pay more trading costs.
- **Main baseline result.** Before trading costs, individual investors earn returns close to a broad market index. After commissions and bid–ask spreads, they underperform.
- **Main economic magnitude.** The average household earns an annual net return of about 16.4%, while the market earns about 17.9%. The average household turns over roughly 75% of its common-stock portfolio each year.
- **Main style fact.** Households tilt toward high-beta, small-cap stocks, with a less pronounced tilt toward value stocks. These tilts help rather than hurt during the sample period, so poor net performance is not mainly a bad-style story.
- **Main overconfidence result.** The highest-turnover households do not earn higher gross returns. Instead, their net performance collapses: the top turnover quintile earns a net annualized return of about 11.4%, while the lowest turnover quintile earns about 18.5%.
- **Main interpretation.** The evidence is consistent with overconfidence models in which investors overestimate the value of their private information, trade too much, and lose money after costs.
- **Alternative explanations.** Liquidity needs, rebalancing, taxes, and entertainment can explain some trading, but the authors argue that they cannot plausibly explain the very high turnover and the strong negative relation between turnover and net performance.
- **Economic takeaway.** The paper’s message is deliberately simple: for the typical individual investor in this sample, active trading is hazardous to wealth because trading costs consume returns without generating offsetting stock-selection gains.

2 Data and measurement (key objects)

2.1 Household account data

The primary data come from a large discount brokerage firm. The original sample contains about 78,000 households from January 1991 through December 1996. The analysis focuses on 66,465 households that hold common stocks during at least one month.

Interpretation. The empirical setting is unusually powerful because it observes both the holdings and the trades of households, rather than inferring investor behavior from aggregate volume or survey data.

2.2 Sample restrictions and household coverage

- The authors exclude mutual funds, American Depositary Receipts, warrants, and options.
- The analysis focuses on directly held common stocks.
- The sampled households have, on average, about two accounts.
- The mean household holds about 4.3 stocks worth \$47,334; the median household holds about 2.61 stocks worth \$16,210.
- In December 1996, the sampled households hold more than \$4.5 billion in common stocks.

Interpretation. These are not simulated portfolios. They are actual household portfolios, often poorly diversified and small enough that proportional trading costs can be economically large.

2.3 Trade-size and transaction-cost measures

For each trade, the authors observe transaction price, commission, and trade value. They estimate two cost components:

- **Commissions:** the dollar commission divided by the principal value of the trade.
- **Bid–ask spread / execution cost:** the transaction price compared with the same-day CRSP closing price, with the sign adjusted for purchases and sales.

A convenient way to write the spread-cost idea is:

$$\text{spread}_{it}^{buy} \approx \frac{P_{it}^{buy}}{P_{it}^{close}} - 1, \quad (1)$$

$$\text{spread}_{it}^{sell} \approx \frac{P_{it}^{close}}{P_{it}^{sell}} - 1. \quad (2)$$

Interpretation. The spread measure captures not only the quoted bid–ask spread but also any execution cost and market impact embedded in the gap between the trade price and the closing price.

2.4 Turnover

The paper measures household trading intensity using purchase and sale turnover. Monthly purchase turnover is the value of shares purchased in month $t - 1$, scaled by the beginning-of-month market value of shares held in month t . Sales turnover is defined analogously.

Mean monthly turnover is the average of purchase turnover and sales turnover:

$$\text{Turnover}_{ht} = \frac{1}{2} (\text{Purchase Turnover}_{ht} + \text{Sales Turnover}_{ht}). \quad (3)$$

Interpretation. Turnover is the main behavioral variable. It summarizes how aggressively each household changes its stock portfolio.

2.5 Gross household returns

For household h , the gross monthly portfolio return is the value-weighted return on the stocks held at the beginning of the month:

$$R_{ht}^{gr} = \sum_{i=1}^{S_{ht}} p_{iht} R_{it}^{gr}, \quad (4)$$

where p_{iht} is the beginning-of-month portfolio weight of stock i , R_{it}^{gr} is the CRSP gross monthly return of stock i , and S_{ht} is the number of stocks held by household h .

Interpretation. Gross returns measure what the household earns from the stocks it holds before paying trading costs.

2.6 Net household returns

The net return adjusts for commissions and spread costs associated with purchases and sales. A compact version of the security-level adjustment is:

$$1 + R_{iht}^{net} = (1 + R_{it}^{gr}) \frac{1 - c_{iht}^s}{1 + c_{ih,t-1}^b}, \quad (5)$$

where c^s is the sale cost and c^b is the purchase cost.

The household-level net return is then:

$$R_{ht}^{net} = \sum_{i=1}^{S_{ht}} p_{iht} R_{iht}^{net}. \quad (6)$$

Interpretation. This adjustment is central. The paper's main finding is not that households pick catastrophically bad stocks before costs; it is that trading costs turn ordinary gross performance into poor net performance.

2.7 Aggregate versus average household returns

The authors report two portfolio concepts.

- **Aggregate household return:**

$$R_t^{AG} = \sum_{h=1}^{n_t} X_{ht} R_{ht}, \quad (7)$$

where X_{ht} is household h 's share of total household stock wealth.

- **Average household return:**

$$R_t^H = \frac{1}{n_t} \sum_{h=1}^{n_t} R_{ht}. \quad (8)$$

Interpretation. Aggregate returns weight dollars; average-household returns weight people. Reporting both shows whether results are driven by large accounts or are typical across households.

3 Models and empirical specifications (all equations + interpretation)

3.1 Own-benchmark abnormal return

The own-benchmark return is the return the household would have earned by holding its beginning-of-year portfolio throughout the year. The abnormal return is:

$$AR_{ht}^{own} = R_{ht} - R_{ht}^{benchmark}. \quad (9)$$

Interpretation. This is the cleanest measure of the effect of trading because each household chooses its own benchmark. If a household did not trade, its own-benchmark abnormal return would be zero.

3.2 Market-adjusted abnormal return

The market-adjusted return subtracts a broad value-weighted market index:

$$AR_t^{mkt} = R_t - R_{mt}. \quad (10)$$

Interpretation. This asks whether household portfolios outperform the market, but it does not adjust for their small-stock and value-stock tilts.

3.3 CAPM alpha

The CAPM specification is:

$$R_t - R_{ft} = \alpha + \beta(R_{mt} - R_{ft}) + \varepsilon_t. \quad (11)$$

Interpretation. The intercept α is Jensen's alpha. A negative α means the household portfolio underperforms after adjusting for market exposure.

3.4 Fama–French three-factor alpha

The preferred factor specification is:

$$R_t - R_{ft} = \alpha + \beta(R_{mt} - R_{ft}) + sSMB_t + hHML_t + \varepsilon_t. \quad (12)$$

Here SMB_t is the small-minus-big factor and HML_t is the high-minus-low book-to-market factor.

Interpretation. This model matters because households tilt toward small stocks. The Fama–French alpha asks whether the portfolios outperform after controlling for market, size, and value exposures.

3.5 Turnover-quintile test of overconfidence

The core behavioral test sorts households into quintiles by mean monthly turnover. Let $q = 1$ denote the lowest-turnover households and $q = 5$ the highest-turnover households. For each quintile, the authors estimate raw returns, own-benchmark abnormal returns, market-adjusted returns, CAPM alphas, and Fama–French alphas.

Interpretation. Rational information-based trading predicts that high-turnover investors should earn higher gross returns to compensate for trading costs. Overconfidence predicts that high-turnover investors will not earn higher gross returns and will earn lower net returns. The data support the overconfidence prediction.

3.6 Momentum-augmented specification

The authors also consider a four-factor specification that adds a price-momentum factor:

$$R_t - R_{ft} = \alpha + \beta(R_{mt} - R_{ft}) + sSMB_t + hHML_t + mMOM_t + \varepsilon_t. \quad (13)$$

Interpretation. Individual investors tend to hold stocks that have recently underperformed. Adding momentum reduces some estimated underperformance, but it does not overturn the main finding that high-turnover households underperform low-turnover households.

3.7 Taxable versus tax-deferred accounts

To evaluate tax-loss selling as an alternative explanation, the paper separately analyzes taxable accounts and tax-deferred accounts such as IRAs and Keoghs.

Interpretation. If tax motives explained most trading, turnover and underperformance should be much weaker in tax-deferred accounts. Instead, turnover remains high and net performance remains poor in both account types.

4 Identification and interpretation (what makes the estimates credible)

4.1 No instrument, but unusually direct measurement

The paper is not an instrumental-variables causal design. Its credibility comes from direct measurement of actual positions, trades, commissions, and transaction prices for a very large set of households.

Interpretation. The paper does not need to infer turnover or costs from aggregate data; it observes them at the household-account level.

4.2 Why the net-return result is hard to dismiss

The central result compares gross and net returns for the same investors. Gross returns are ordinary; net returns are poor. The difference is mechanically tied to observed trading costs.

Interpretation. This makes the main accounting point compelling: trading itself is costly, and the observed stock-selection gains are not large enough to offset those costs.

4.3 Why own-benchmark returns are important

The own-benchmark abnormal return compares the household's actual performance with what it would have earned by not trading its own initial portfolio.

Interpretation. This avoids imposing a controversial asset-pricing model and directly asks whether trading improved performance relative to buy-and-hold.

4.4 Why factor adjustments matter

Households hold high-beta, small-cap, and somewhat value-oriented portfolios. Without factor adjustment, one might confuse factor exposure with skill or poor performance.

Interpretation. The Fama–French results show that net underperformance survives reasonable adjustment for the style of stocks households choose.

4.5 Overconfidence versus rational information trading

The turnover-quintile design is the paper's main behavioral comparison.

- **Rational information story:** investors trade because private information is valuable; high-turnover investors should earn higher gross returns.
- **Overconfidence story:** investors overestimate information quality; high-turnover investors trade more, but do not earn enough gross return to cover costs.

Interpretation. The absence of higher gross returns for active traders is what makes the result especially damaging for the rational information story.

4.6 Alternative explanations

- **Liquidity.** Some trading may finance liquidity shocks, but mutual funds would allow lower-cost liquidity management.
- **Risk rebalancing.** Some rebalancing is rational, but the typical household holds only a few stocks, making 75% annual turnover difficult to justify as precision rebalancing.
- **Taxes.** Tax-loss selling can explain some trades, especially in December, but not high turnover and underperformance in tax-deferred accounts.
- **Entertainment or gambling.** Some investors may enjoy trading, but the authors favor overconfidence because investors generally do not expect their trades to lower wealth.

4.7 Main limitation

The sample comes from one large discount broker and only observes assets held at that broker. Households may also hold mutual funds, retirement assets, or accounts elsewhere.

Interpretation. The paper is strongest about the performance of directly held common stocks in observed brokerage accounts. It is less informative about the full household balance sheet.

5 Tables and figures

5.1 Figure 1: Monthly turnover and annual performance

Read:

- Gross annualized returns are similar across turnover quintiles.
- Net annualized returns fall sharply as turnover rises.
- The highest-turnover quintile has monthly turnover above 20%, implying more than 250% annualized turnover.
- The average household trades frequently and earns a lower net return than the passive benchmark.

Economic message. The central fact is the wedge between gross and net performance. High-turnover investors do not appear to discover better stocks; they mostly pay more trading costs.

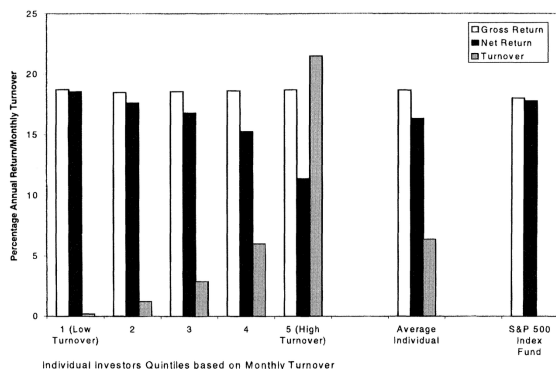


Figure 1. Monthly turnover and annual performance of individual investors. The white bar (black bar) represents the gross (net) annualized geometric mean return for February 1991 through January 1997 for individual investor quintiles based on monthly turnover, the average individual investor, and the S&P 500. The net return on the S&P 500 Index Fund is that earned by the Vanguard Index 500. The gray bar represents the monthly turnover.

Table I
Descriptive Statistics on Trade Size, Trade Price, Transaction Costs, and Turnover

The sample is account records for 66,465 households at a large discount brokerage firm from January 1991 to December 1996. Spread is calculated as the transaction price divided by the closing price on the day of the transaction minus one (and then multiplied by minus one for purchases). Commission is calculated as the commission paid divided by the value of the trade. Monthly turnover is the beginning-of-month market value of shares purchased in month $t-1$ (or sold in month t) divided by the total beginning-of-month market value of shares held in month t . Trade-weighted spread and commission are averages weighted by trade size. Aggregate turnover is the aggregate value of sales (or purchases) divided by the aggregate value of positions held during our sample period.

	Mean	25th Percentile	Median	75th Percentile	Standard Deviation	No. of Obs.
Panel A: Purchases						
Trade size (\$)	11,205	2,513	4,988	10,500	32,179	1,082,107
Price/share	31.06	11.00	23.00	40.00	117.82	1,082,107
Monthly turnover (%)	6.49	0.54	2.67	7.08	11.89	66,465
Commission (%) [*]	1.58	0.78	1.29	2.10	1.45	966,492
Spread (%)	0.31					1,028,087
Panel B: Sales						
Trade size (\$)	13,707	2,688	5,738	13,000	38,275	887,594
Price/share	31.22	12.00	24.00	41.00	113.03	887,594
Monthly turnover (%)	6.23	0.39	2.58	6.95	11.36	66,465
Commission (%) [*]	1.45	0.70	1.16	1.91	1.06	785,206
Spread (%)	0.69					845,644
Panel C: Trade-Weighted and Aggregate Purchases						
Aggregate monthly turnover (%)	6.05					
Trade-weighted commission (%)	0.77			Not Applicable		
Trade-weighted spread (%)	0.27					
Panel D: Trade-Weighted Sales						
Aggregate monthly turnover (%)	6.06					
Trade-weighted commission (%)	0.66			Not applicable		
Trade-weighted spread (%)	0.61					

^{*}Commissions are calculated based on trades in excess of \$1,000. Including smaller trades results in a mean buy (sale) commission of 2.09 (3.07) percent.

5.2 Table I: Trade size, transaction costs, and turnover

Read:

- Purchases and sales are both numerous: more than one million purchases and almost 0.9 million sales.
- Average purchase turnover is 6.49% per month; average sale turnover is 6.23% per month.
- Average commissions on trades above \$1,000 are about 1.58% for purchases and 1.45% for sales.
- Trade-weighted commissions are lower, but still economically meaningful.

Economic message. Trading is not a tiny friction. The typical household trades often enough, and pays high enough proportional costs, for trading to have a first-order effect on performance.

5.3 Table II: Full-sample abnormal performance

Read:

- Gross performance is close to ordinary market performance.
- Own-benchmark gross abnormal returns are negative, suggesting trading lowers performance even before costs.
- Net own-benchmark abnormal returns are strongly negative.
- Net Fama–French alphas are negative, especially for the average household.

Economic message. Once transaction costs and stock-style exposures are accounted for, the average household underperforms. The poor result is not explained by high-beta or small-stock exposure.

5.4 Table V: Turnover quintiles

Read:

- Mean monthly turnover rises from 0.19% in the lowest quintile to 21.49% in the highest quintile.
- High-turnover households have much stronger exposure to small stocks and higher market beta.
- Gross returns do not rise enough to compensate for more trading.
- Net Fama–French alpha falls from about -0.061% per month in the lowest-turnover quintile to about -0.864% per month in the highest-turnover quintile.

Economic message. This is the key behavioral test. The investors who trade most are not rewarded with superior gross performance; they suffer much worse net performance.

Table II

Summary of the Percentage Monthly Abnormal Return Measures for the Average Household and Aggregate Household

Returns are based on month-end position statements for 66,465 households at a large discount brokerage firm from January 1991 to December 1996. Panel A (Panel C) presents results for the gross (net) return on a portfolio that mimics the aggregate investment of all households. Panel B (Panel D) presents results for the gross (net) return on a portfolio that mimics the investment of the average household. Own-benchmark abnormal return is the return on the household portfolio minus the return on the portfolio the household held at the end of the previous January. Market-adjusted return is the return on the household portfolio less the return on a value-weighted NYSE/AMEX/Nasdaq index. CAPM is the results from a time-series regression of the household excess return on the market excess return ($R_{mt} - R_{ft}$). Fama-French three-factor is the results from time-series regression of household excess return on the market excess return, a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t). p -values are presented in parentheses.

	Excess Return	Coefficient Estimate on:			Adjusted R^2
		$(R_{mt} - R_{ft})$	HML_t	SMB_t	
Panel A: Gross Percentage Monthly Returns in Aggregate					
Own-benchmark abnormal return	-0.049** (0.013)				
Market-adjusted return	0.038 (0.723)				
CAPM	-0.067 (0.543)	1.100*** (0.007)			92.9
Fama-French three-factor	-0.076 (0.357)	1.082*** (0.005)	-0.035 (0.324)	0.231*** (0.000)	96.3
Panel B: Gross Percentage Monthly Returns for the Average Household					
Own-benchmark abnormal return	-0.048** (0.010)				
Market-adjusted return	0.078 (0.672)				
CAPM	-0.014 (0.944)	1.087 (0.177)			80.3
Fama-French three-factor	-0.154 (0.205)	1.120*** (0.005)	0.140*** (0.008)	0.516*** (0.000)	93.0
Panel C: Net Percentage Monthly Returns in Aggregate					
Own-benchmark abnormal return	-0.155*** (0.000)				
Market-adjusted return	-0.073 (0.496)				
CAPM	-0.175 (0.113)	1.096*** (0.009)			93.0
Fama-French three-factor	-0.180** (0.031)	1.077*** (0.009)	-0.040 (0.251)	0.225*** (0.000)	96.3
Panel D: Net Percentage Monthly Returns for the Average Household					
Own-benchmark abnormal return	-0.194*** (0.000)				
Market-adjusted return	-0.090 (0.621)				
CAPM	-0.177 (0.360)	1.082 (0.194)			80.7
Fama-French three-factor	-0.311** (0.011)	1.113*** (0.008)	0.131** (0.012)	0.506*** (0.000)	93.0

***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively (two-tailed). The null hypothesis for beta (the coefficient estimate on the market excess return) is $H_0: \beta = 1$.

Table IV

Cross-Sectional Distribution of Percentage Monthly Gross and Net Market-Adjusted Household Returns

The sample is account records for 66,465 households at a large discount brokerage firm from January 1991 to December 1996. Households with position statements in 12 or fewer months are omitted from this analysis. Though the median values are virtually identical when these households are included, more extreme values are observed.

	Gross Monthly Market-Adjusted Return (%)	Net Monthly Market-Adjusted Return (%)
Minimum	-19.46	-20.85
1st percentile	-4.32	-4.86
5th percentile	-2.12	-2.45
10th percentile	-1.34	-1.60
25th percentile	-0.57	-0.73
Median	-0.01	-0.14
75th percentile	0.66	0.50
90th percentile	1.62	1.40
95th percentile	2.41	2.15
99th percentile	4.86	4.44
Maximum	48.53	48.35
Total households	62,439	62,439
Percentage > 0	49.3%***	43.4%***
Binomial Z-statistic	-3.38	-33.13

*** indicates significant difference from 50 percent at the 1% level.

5.5 Table VI: Taxable and tax-deferred accounts

Read:

- Taxable accounts have higher monthly turnover than tax-deferred accounts, but turnover is high in both.
- Gross returns are similar across account types.
- Net performance is negative after factor adjustment in both taxable and tax-deferred accounts.
- Tax motives therefore cannot fully explain the trading-performance relation.

Economic message. Tax-loss selling is not the whole story. The persistence of high turnover and underperformance in tax-deferred accounts supports a behavioral trading interpretation.

Table V
Descriptive Statistics, Gross Returns, and Net Returns for
Household Quintiles Formed on Mean Turnover

The sample is account records for 66,465 households at a large discount brokerage firm from January 1991 to December 1996. Households are sorted into quintiles based on monthly turnover (the average of sales and purchase turnover) during our sample period. Quintile 1 contains households with the lowest turnover, quintile 5 contains households with the highest. Beginning position value is the market value of common stocks held in the first month that the household appears during our sample period. Mean monthly turnover is the average of sales and purchase turnover. Coefficient estimates are those from a time-series regression of the gross average household excess return on the market excess return ($R_{mt} - R_{ft}$), (HML_t), and a zero-investment size portfolio (SMB_t). Raw return is the average monthly return for the average household. Own-benchmark abnormal return is the return on the household portfolio minus the return on the portfolio the household held at the end of the previous January. Market-adjusted return is the return on the household portfolio less the return on a value-weighted NYSE/AMEX/Nasdaq index. CAPM intercept is the estimated intercept from a time-series regression of the household excess return on the market excess return ($R_{mt} - R_{ft}$). Fama-French intercept is the estimated intercept from time-series regressions of household excess return on the market excess return, a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t). p -values are presented in parentheses.

	Quintile					Difference: High - Low
	(1 Low)	2	3	4	(5 High)	
Panel A: Descriptive Statistics						
Mean monthly turnover (%)	0.19	1.24	2.89	5.98	21.49	N.A.
Mean beginning position value	54,169	26,046	22,945	19,102	21,560	-12,609*** (0.000)
Coefficient estimate on ($R_{mt} - R_{ft}$)	1.03 (0.199)	1.06* (0.090)	1.11** (0.015)	1.18*** (0.002)	1.29*** (0.000)	0.26*** (0.000)
HML_t	0.20*** (0.000)	0.10*** (0.012)	0.13** (0.020)	0.13* (0.065)	0.12 (0.195)	-0.08 (0.333)
SMB_t	0.24*** (0.000)	0.29*** (0.000)	0.51*** (0.000)	0.72*** (0.000)	1.02*** (0.000)	0.78*** (0.000)
Adjusted R^2	96.1	94.7	92.2	90.4	87.6	71.8
Panel B: Gross Average Household Percentage Monthly Return						
Raw return	1.483	1.472	1.489	1.511	1.548	0.065
Own-benchmark abnormal return	-0.009 (0.156)	-0.026* (0.064)	-0.052** (0.014)	-0.079*** (0.007)	-0.096* (0.093)	-0.087 (0.116)
Market-adjusted return	0.063 (0.534)	0.052 (0.660)	0.069 (0.710)	0.091 (0.726)	0.128 (0.728)	0.065 (0.832)
CAPM intercept	0.090 (0.409)	0.022 (0.865)	-0.015 (0.936)	-0.078 (0.774)	-0.167 (0.663)	-0.257 (0.407)
Fama-French intercept	-0.048 (0.526)	-0.072 (0.448)	-0.149 (0.242)	-0.237 (0.143)	-0.359 (0.104)	-0.311* (0.086)
Panel C: Net Average Household Percentage Monthly Return						
Raw return	1.470	1.411	1.361	1.267	1.009	-0.460
Own-benchmark abnormal return	-0.021*** (0.000)	-0.078*** (0.000)	-0.167*** (0.000)	-0.300*** (0.000)	-0.587*** (0.000)	-0.566*** (0.000)
Market-adjusted return	0.050 (0.625)	-0.009 (0.937)	-0.059 (0.749)	-0.153 (0.547)	-0.411 (0.253)	-0.460 (0.124)
CAPM intercept	0.077 (0.480)	-0.038 (0.764)	-0.140 (0.474)	-0.314 (0.242)	-0.692* (0.066)	-0.768** (0.012)
Fama-French intercept	-0.061 (0.422)	-0.130 (0.172)	-0.269** (0.037)	-0.464*** (0.005)	-0.864*** (0.000)	-0.803*** (0.000)

***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively (two-tailed). The null hypothesis for beta (the coefficient estimate on the market excess return) is $H_0: \beta = 1$ except in the difference column, where the null hypothesis is $H_0: \beta = 0$.

6 Short summary

- The paper studies 66,465 households with direct common-stock investments at a discount broker.
- Households trade frequently: the average household turns over about 75% of its common-stock portfolio per year.
- Gross returns are close to market returns, but net returns are poor after commissions and bid–ask costs.
- The average household tilts toward high-beta, small-cap stocks; the average household also has some value tilt.
- Own-benchmark abnormal returns show that households would have done better by holding their beginning-of-year portfolios.
- Turnover is the crucial cross-sectional variable: high-turnover households do not earn higher gross returns, but earn much lower net returns.
- Liquidity, rebalancing, taxes, and entertainment may explain part of trading, but they do not explain the full pattern.
- The evidence fits overconfidence models in which investors overestimate their information and trade too much.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

Barber and Odean show that individual investors who trade common stocks directly earn ordinary gross returns but poor net returns. The performance penalty is especially severe among the most active traders. After accounting for commissions, bid–ask spreads, and factor exposures, active trading reduces household wealth rather than increasing it.

7.2 Economic intuition

The intuition is simple. If investors had valuable private information, frequent trading should produce higher gross returns. Instead, active traders earn gross returns similar to other investors but pay much larger trading costs. Overconfidence provides a natural explanation: investors believe their signals are more precise than they really are, trade aggressively, and lose money net of costs.

7.3 Takeaway

The paper’s lasting contribution is to document, using actual household brokerage records, that active trading by individual investors is costly and usually unrewarded. The memorable lesson is also the paper’s title: **trading is hazardous to your wealth.**